

Nonparametric estimation on the circle based on Fejér polynomials

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1 The problem

Nonparametric density and distribution function estimation are well developed for Euclidean data, but circular data require estimators that respect periodicity. Some methods transfer Bernstein polynomial estimators from the unit interval to the circle, although this does not directly use the periodic structure of a circular density.

This paper develops nonparametric density and distribution function estimators on the circle using Fejér polynomials, which are periodic analogues of Bernstein polynomials. It also extends the estimators to Berkson and classical measurement error models.

2 The setup

Let

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} X$$

be circular observations with density f and distribution function F on $[-\pi, \pi)$. The goal is to estimate f and F while respecting the periodic sample space.

The Fejér kernel of order m is

$$\begin{aligned} K_m(s) &= \frac{1}{2\pi} \sum_{k=-m}^m \left(1 - \frac{|k|}{m+1}\right) e^{iks} \\ &= \frac{1}{2\pi} + \frac{1}{\pi} \sum_{k=1}^m \left(1 - \frac{k}{m+1}\right) \cos(ks). \end{aligned}$$

It is continuous, symmetric, periodic, nonnegative, and satisfies

$$\int_{-\pi}^{\pi} K_m(s) ds = 1.$$

The paper also considers two circular measurement error models. In the Berkson model,

$$X = (X^* + \varepsilon) \bmod 2\pi,$$

where X^* is observed and X is the target variable. In the classical model,

$$X^* = (X + \varepsilon) \bmod 2\pi,$$

where X^* is observed and X is latent. The error distribution is assumed to be known.

3 The main idea

For a continuous periodic density f , Fejér's theorem gives the uniform approximation

$$\sigma_m(f; \theta) = \frac{1}{2\pi} \sum_{k=-m}^m \left(1 - \frac{|k|}{m+1}\right) \phi_k e^{ik\theta},$$

where

$$\phi_k = \int_{-\pi}^{\pi} f(t) e^{-ikt} dt$$

is the k -th Fourier coefficient of f .

Replacing ϕ_k by

$$\hat{\phi}_{n,k} = \frac{1}{n} \sum_{j=1}^n e^{-ikX_j}$$

gives the density estimator

$$\hat{f}_{m,n}(\theta) = \frac{1}{2\pi} \sum_{k=-m}^m \left(1 - \frac{|k|}{m+1}\right) \hat{\phi}_{n,k} e^{ik\theta}.$$

Equivalently,

$$\hat{f}_{m,n}(\theta) = \frac{1}{n} \sum_{j=1}^n K_m(\theta - X_j).$$

For distribution function estimation, define

$$\begin{aligned} W_m(\theta) &= \int_{-\pi}^{\theta} K_m(y) dy \\ &= \frac{\theta + \pi}{2\pi} + \frac{1}{\pi} \sum_{k=1}^m \left(1 - \frac{k}{m+1}\right) \frac{\sin(k\theta)}{k}. \end{aligned}$$

Then

$$\hat{F}_{m,n}(\theta) = \frac{1}{n} \sum_{i=1}^n \{W_m(\theta - X_i) - W_m(-\pi - X_i)\}.$$

A circular distribution function depends on the selected origin. For an origin θ_0 , define

$$F^{\theta_0}(\theta) = \int_{\theta_0}^{\theta} f(y) dy.$$

After transforming the data to $[\theta_0, \theta_0 + 2\pi)$, let

$$x_{(1)}^{\theta_0} \leq \dots \leq x_{(n)}^{\theta_0}$$

denote the ordered observations. The origin is selected by minimizing

$$C_n(\theta_0) = \sum_{i=1}^n \frac{i}{n} \left(1 - \frac{i}{n}\right) (x_{(i+1)}^{\theta_0} - x_{(i)}^{\theta_0}),$$

using the circular ordering convention for the final spacing.

For measurement error, let λ_ℓ be the Fourier coefficients of the error density. The Berkson estimator is

$$\hat{f}_{\text{Berk}}(x; m) = \frac{1}{2\pi n} \sum_{j=1}^n \left[1 + 2 \sum_{\ell=1}^m \left(1 - \frac{\ell}{m+1}\right) \lambda_\ell \cos\{\ell(x - X_j^*)\} \right].$$

In the classical model, the corresponding estimator is

$$\hat{f}_{\text{class}}(x; m) = \frac{1}{2\pi n} \sum_{j=1}^n \left[1 + 2 \sum_{\ell=1}^m \frac{1 - \ell/(m+1)}{\lambda_{\ell}} \cos\{\ell(x - X_j^*)\} \right].$$

Thus, the Berkson correction multiplies the Fourier coefficients by the error coefficients, whereas the classical correction divides by them.

4 The main results

The paper establishes uniform strong consistency of the Fejér density estimator by combining Fejér’s uniform approximation theorem with convergence of the empirical Fourier coefficients.

It also derives a MISE expansion. If f is not the circular uniform density, the asymptotically optimal order is written as

$$m_{\text{opt}} = \{6\pi\theta_1(f)\}^{1/3} n^{1/3}$$

in the notation of the paper.

The Fejér estimator is extended to both Berkson and classical measurement error models. In the Berkson case, the smoothing order can be selected as in the error-free setting. In the classical case, division by the error Fourier coefficients increases the variance of the estimator.

5 My takeaways

- How does the Fejér weighting $1 - |k|/(m+1)$ ensure that the density estimator remains nonnegative, unlike an ordinary truncated Fourier series?
- The density itself does not depend on an origin, but a circular distribution function does. How strongly can the selected origin affect finite-sample CDF estimation?
- Why does the Berkson correction multiply by λ_{ℓ} , whereas the classical correction divides by λ_{ℓ} , and how does this difference affect the variance?

References

- [1] B. Klar, B. Milošević, and M. Obradović. Nonparametric estimation on the circle based on Fejér polynomials. *arXiv preprint arXiv:2411.19138*, 2024. <https://doi.org/10.48550/arXiv.2411.19138>.